SILVER OAK UNIVERSITY
EDUCATION TO INNOVATION

A PROJECT DISSERTATION / REPORT ON

ISMS COMPLIANCE ASSESSMENT PLATFORM

An Enterprise-Grade Multi-Tenant ISO/IEC 27001:2022 Compliance Automation, Risk Management & SoA Generation SaaS System

Submitted in partial fulfilment of the requirements for the award of the degree of
Bachelor of Computer Applications (Honours)

SUBMITTED BY:
Kaswan Sangeeta Bhimsingh
Roll No: 17
BCA (Honours)

AFFILIATED TO:
Silver Oak University
Ahmedabad, Gujarat

UNDER THE VALUED GUIDANCE OF
Prof. Harsh Barot
Assistant Professor, Department of Computer Science
Silver Oak College of Computer Applications
Ahmedabad, Gujarat, India
Academic Year 2026–27

SILVER OAK COLLEGE OF COMPUTER APPLICATIONS
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CERTIFICATE

This is to certify that the project report entitled “ISMS Compliance Assessment Platform” is a bonafide record of the technical development work carried out by Kaswan Sangeeta Bhimsingh (Roll No: 17) and submitted to Silver Oak University in partial fulfillment of the requirements for the award of the degree of Bachelor of Computer Applications (Honors) during the academic year 2026–27.

The system has been architected under strict multi-tenant isolation, enterprise repository-service patterns, and implements the full ISO/IEC 27001:2022 ISMS framework. To the best of our knowledge, this work is original, carried out under the direct supervision of Prof. Harsh Barot, and has not been submitted elsewhere for any academic recognition.

_____	_____
Prof. Harsh Barot	Head of Institution / Principal
Project Guide	Silver Oak College of Computer Applications
Silver Oak College of Computer Applications	

Place: Ahmedabad, Gujarat
Date: 29-10-2026

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Acknowledgement

Building an enterprise-grade Information Security Management System (ISMS) compliance SaaS platform has been an enriching experience that bridged enterprise software architecture with rigorous information security standards. I express my profound gratitude to all who facilitated this journey.

I am deeply thankful to Silver Oak College of Computer Applications, Ahmedabad, for providing state-of-the-art computational infrastructure, server environments, and an academically motivating atmosphere.

I record my heartfelt gratitude to the leadership and faculty of Silver Oak College of Computer Applications, for their continuous encouragement and support in facilitating research and development projects.

I am immensely indebted to my project guide, Prof. Harsh Barot, for his technical guidance, architectural critique, and rigorous review of the multi-tenancy isolation and ISO/IEC 27001:2022 control mapping.

I also extend my sincere thanks to Silver Oak University for prescribing an academic structure that encourages the creation of high-impact, industry-grade software solutions.

Kaswan Sangeeta Bhimsingh
BCA (Honours) — Roll No: 17
Silver Oak College of Computer Applications

5. Project Overview, Objectives & Architectural Philosophy

SYSTEM PROFILE & TECHNOLOGY SPECIFICATION

Platform Name: ISMS Compliance Assessment Platform

Standard Supported: ISO/IEC 27001:2022 (Information Security, Cybersecurity & Privacy Protection)

Primary Technology Stack: Laravel 11.x, PHP 8.2+, MySQL 8.0, Blade Templates, Tailwind CSS 4, Alpine.js, Chart.js

Architectural Invariants: Strict Service Layer, Repository Pattern, DTOs, TenantScope Global Isolation, Policy-based RBAC

Live Demonstration Environment: https://ismsaudit.infinityfree.io

Achieving and maintaining ISO/IEC 27001:2022 compliance is traditionally a tedious, fragmented process involving dispersed spreadsheets, disconnected audit notes, and manual Statement of Applicability (SoA) tracking. The ISMS Compliance Assessment Platform automates this lifecycle by offering a centralized, multi-tenant SaaS environment that enforces data isolation, risk calculations (5x5 matrix), evidence attachments, and live analytics.

5.1 Non-Negotiable Enterprise Architectural Rules

- **Zero Business Logic in Controllers:** Controllers strictly handle HTTP validation via FormRequests, invoke dedicated Services, and return View/JSON responses. All domain calculations reside in app/Services.
- **Data Access Decoupling via Repositories:** Direct Eloquent queries are abstracted behind Repository interfaces (app/Repositories/Contracts), facilitating decoupled unit testing and database independence.
- **Immutable Data Transfer Objects (DTOs):** Controller-to-service communication is strictly mediated using strongly-typed PHP 8.2 DTOs (app/DTOs), preventing loose array tampering.
- **Global Scope Multi-Tenancy:** Tenant isolation is enforced at the database level via TenantScope. Every query on tenant entities is automatically scoped with WHERE tenant_id = :current_tenant.

6. PHASE 1: Foundation, Multi-Tenancy Core & ISO 27001 Seeding

> **PHASE 1 MILESTONES — COMPLETED AUGUST 17, 2026**
>
> **Objectives:** Establish database schemas, global tenant scopes, Spatie RBAC, and seed full ISO 27001:2022 structure.
>
> **Verification Status:** 29 / 29 PHPUnit Feature & Unit Tests Passing (100% Green).

Phase 1 established the foundation of the platform. A shared-database, shared-schema multi-tenant architecture was engineered. The global scope automatically resolves the tenant context from the session of the authenticated user, guaranteeing complete tenant segregation.

Architecture Specification / Implementation — app/Scopes/TenantScope.php — Automatic Query Filtering

```
class TenantScope implements Scope
{
    public function apply(Builder $builder, Model $model): void
    {
        if (auth()->check() && auth()->user()->current_tenant_id) {
            if (Schema::hasColumn($model->getTable(), 'tenant_id')) {
                $builder->where($model->getTable() . '.tenant_id', auth()->user()-
>current_tenant_id);
            }
        }
    }
}
```

Verified Terminal Output / System Assertion:

```
Verified: Invariant prevents cross-tenant access even if raw query lacks explicit where clause.
```

6.1 Full ISO/IEC 27001:2022 Control Seeder Implementation

A comprehensive seeder populated all 11 domains (4 main clauses 4–10 and 4 Annex A themes A.5 to A.8) and 115 controls (93 Annex A controls + 14 main clause checkpoints + 8 extended controls) with categorized control types (preventive, detective, corrective).

ISO 27001:2022 Theme / Clause	Domain Code	Control Count	Security Category
Organizational Controls	Clause 4–10 & A.5	45 Controls	Organizational Governance
People Controls	Annex A.6	8 Controls	Human Resource Security
Physical Controls	Annex A.7	14 Controls	Physical & Environmental
Technological Controls	Annex A.8	48 Controls	System, Network & Data Security

TOTAL SEEDED AUDIT CONTROLS	11 Domains	115 Controls	Complete 2022 Standard Coverage

7. PHASE 2: Assessment Engine, Maturity Scoring & Evaluation Workflow

> **PHASE 2 MILESTONES — COMPLETED AUGUST 19, 2026**
>
> **Objectives:** Multi-step assessment wizard, domain-by-domain evaluation, maturity scoring (Level 0-5), and ComplianceCalculator.
>
> **Verification Status:** 37 / 37 Feature Tests Passing (93 assertions).

Phase 2 introduced the core Assessment Engine. Auditors and assessors can instantiate assessments, assign assessors via the assessment_team pivot, and systematically evaluate controls across five compliance states:

- **Compliant (100%):** Full adherence to the control statement with verified supporting artifact.
- **Partially Compliant (50%):** Policy documented or ad-hoc process active, but lacks continuous enforcement.
- **Non-Compliant (0%):** Control statement not addressed, creating an identified audit gap.
- **Not Applicable:** Control excluded from ISMS scope with mandatory justification recorded.

Architecture Specification / Implementation — app/Services/Compliance/ComplianceCalculator.php — Weighted Compliance Formula

```
public function calculateOverallScore(Assessment $assessment): float
{
    $responses = $this->responseRepo->getApplicableForAssessment($assessment->id);
    if ($responses->isEmpty()) return 0.0;

    $totalWeight = $responses->count() * 100;
    $earnedScore = $responses->sum(function ($r) {
        return match($r->status) {
            'compliant' => 100,
            'partially_compliant' => 50,
            default => 0
        };
    });
    return round(($earnedScore / $totalWeight) * 100, 2);
}
```

Verified Terminal Output / System Assertion:

```
Score Range: 0.00% to 100.00% dynamic calculation per tenant assessment.
```

[Verified Module Screenshot: docs/screenshots/assessment_conduct_screen.png]

Figure 2.1: *Assessment Evaluation Workspace: Evaluating Annex A.5 Organizational Controls with live maturity scoring.*

8. PHASE 3: Information Asset Register & Risk Assessment Engine

> **PHASE 3 MILESTONES — COMPLETED AUGUST 28, 2026**
>
> **Objectives:** Asset Register with CIA valuation, 5x5 Likelihood x Impact scoring, and clickable Heatmap visualization.
>
> **Verification Status:** 49 / 49 Feature Tests Passing (112 assertions).

Clause 6.1.2 of ISO/IEC 27001 requires an information security risk assessment process. Phase 3 engineered an Asset Inventory supporting hardware, software, data, people, and facilities. Each asset receives CIA (Confidentiality, Integrity, Availability) ratings on a 1–5 scale to quantify critical asset value.

8.1 5x5 Risk Matrix & Heatmap Topology

Score Range	Risk Severity	Hex Color Code	Automated Action Required
Score 1 – 4	Low Risk	#10B981 (Green)	Accept risk; periodic annual monitoring
Score 5 – 9	Medium Risk	#FBBF24 (Yellow)	Mitigate with existing Annex A controls
Score 10 – 16	High Risk	#F97316 (Orange)	Mandatory Treatment Plan with assigned owner
Score 17 – 25	Critical Risk	#EF4444 (Red)	Immediate management escalation; mitigation required

[Verified Module Screenshot: docs/screenshots/risk_heatmap_5x5.png]

Figure 3.1: *Interactive 5x5 Risk Heatmap matrix rendering live risk distribution across Likelihood and Impact axes.*

9. PHASE 4: Statement of Applicability (SoA) & Version Snapshot Engine

> **PHASE 4 MILESTONES — COMPLETED SEPTEMBER 02, 2026**
>
> **Objectives:** One-click SoA generation, regulatory justifications, multi-tier approvals, and immutable JSON version snapshots.
>
> **Verification Status:** 54 / 54 Feature Tests Passing (128 assertions).

The Statement of Applicability (SoA) is the mandatory document required by ISO 27001 Clause 6.1.3(d). Phase 4 delivered an engine that transforms raw assessment data into formal SoA records, tracks justifications for inclusion or exclusion, and archives version snapshots in `soa_versions`.

Architecture Specification / Implementation — app/Services/Compliance/StatementOfApplicabilityService.php — Immutable Versioning

```
public function createVersionSnapshot(int $assessmentId, string $versionString, int $userId): SoAVersion
{
    $entries = $this->soaRepo->getByAssessmentId($assessmentId);
    return $this->versionRepo->create([
        'assessment_id' => $assessmentId,
        'version'       => $versionString,
        'data_snapshot' => json_encode($entries->toArray()),
        'created_by'    => $userId
    ]);
}
```

Verified Terminal Output / System Assertion:

```
Audit Trail: Generates immutable JSON payload for external certification auditor review.
```

[Verified Module Screenshot: docs/screenshots/soa_version_management.png]

Figure 4.1: *Statement of Applicability (SoA) workspace showing approved controls and side-by-side version comparison.*

10. PHASES 5, 6 & 7: DMS, Analytics Dashboard & Enterprise Queues

INTEGRATED ENGINE SPECIFICATIONS

Phase 5 (DMS): Document lifecycle (Draft → Under Review → Approved), file versioning, and direct control-linking.

Phase 6 (Analytics): Executive KPI cards, Chart.js trend tracking, automated snapshots, and multi-format exports (PDF/Excel/CSV).

Phase 7 (Enterprise): Four risk treatment strategies, automated email + in-app notification center, and scheduled daemon cron jobs.

In Phase 5, an ISMS Document Management System (DMS) was integrated with direct mapping to Annex A controls via `document_control_map`. Phase 6 developed an Executive Dashboard powered by Chart.js, aggregating compliance posture, domain breakdown, and 12-month historical trends.

Phase 7 completed the operational workflow by introducing risk treatment strategies (Mitigate, Accept, Transfer, Avoid), interactive notification centers, and three automated queue workers configured in `routes/console.php`:

- **Daily Compliance Snapshot Job (02:00):** Captures daily organizational compliance score in report_snapshots.
- **Treatment Due Reminders Job (08:00):** Dispatches automated notification emails for pending risk mitigations.
- **Snapshot Retention Pruning Job (Sun 03:00):** Prunes historical compliance metrics exceeding the 24-month retention window.

Report Type	Primary Target Audience	Supported Formats	Data Source Model
Executive Summary	CISO, Board of Directors	PDF, Excel, CSV	DashboardService KPI Aggregator
Gap Analysis Report	Internal Lead Auditor	PDF, Excel, CSV	Non-compliant Assessment Responses
Risk Register Report	Risk & Security Committee	PDF, Excel, CSV	Risk & Treatment Plan Entities
Statement of Applicability	Certification Body (Auditor)	PDF, Excel, XLSX	SoAEntry & SoAVersion Tables
Document Compliance	Compliance Manager	CSV, PDF	Documents & Review Dates

[Verified Module Screenshot: docs/screenshots/executive_analytics_dashboard.png]

Figure 5.1: *Executive Analytics Dashboard: Real-time Compliance Score, Domain Chart & Risk Distribution.*

11. PHASE 8: Enterprise UI/UX Overhaul, Security Hardening & Deployment

> **PRODUCTION READINESS & REFACTORING REVIEW**
>
> **Deployment Target:** InfinityFree Cloud Hosting (Production URL: https://ismsaudit.infinityfree.io)
>
> **Security Interventions:** Policy-based RBAC enforcement, form nesting repairs, IDOR blocking, and custom error boundaries.
>
> **UI Paradigm:** B2B SaaS Aesthetic inspired by Vanta & Drata, built on Tailwind CSS 4 and Alpine.js.

Phase 8 transformed the platform from a functional prototype into a polished, commercial-grade SaaS product. Critical architectural and security vulnerabilities were resolved during this sprint:

11.1 Security Hardening & Vulnerability Remediation

- **RBAC Policy Enforcement:** Implemented AssetPolicy and RiskPolicy. Viewer roles attempting edit or delete operations are blocked with HTTP 403 Forbidden.
- **Form Nesting DOM Bug Correction:** Resolved an issue where HTML <form> delete actions nested inside update forms triggered unintended deletes. Extracted action handlers using explicit HTML5 form attributes.
- **Route Protection & IDOR Mitigation:** Secured `/reports` and `/soa` behind `tenant.selected` middleware; cross-tenant IDOR tampering consistently triggers HTTP 404.
- **Custom HTTP Error Boundaries:** Replaced default framework errors with bespoke Tailwind error pages for 403, 404, 419, 500, and 503 statuses.

Architecture Specification / Implementation — app/Policies/RiskPolicy.php — Role-Based Authorization Guard

```
public function update(User $user, Risk $risk): bool
{
    // Viewers strictly prohibited from altering risk parameters
    if ($user->hasRoleInCurrentTenant('viewer')) {
        return false;
    }
    return $user->current_tenant_id === $risk->tenant_id;
}
```

Verified Terminal Output / System Assertion:

```
Result: Verified via automated tests; unauthorized PUT requests abort with 403 Forbidden.
```

[Verified Module Screenshot: docs/screenshots/login_split_screen.png]

Figure 8.1: *Redesigned Split-Screen Authentication UI featuring Glassmorphism and responsive mobile drawer navigation.*

12. Comprehensive Test Verification Suite & Invariant Matrix

AUTOMATED TEST SUITE STATUS — 100% PASS RATE

Total Executed Tests: 71 Feature & Unit Tests

Total Assertions: 169 Verified Assertions

Framework & Tooling: PHPUnit 10.5, Laravel Feature Testing Engine, SQLite In-Memory Database

Execution Time: ~25.75 seconds

Verification Invariant	Test Case Description & Security Requirement	Status
Tenant Isolation Scope	User from Tenant A cannot read or mutate Tenant B assets/risks	PASS ✓
ISO Control Seeder Count	Exact count of 115 controls (93 Annex A + 14 Main Clause + 8 Ext)	PASS ✓
Compliance Math Formula	Partially compliant scores weight at 50%; non-compliant at 0%	PASS ✓
Risk Auto-Scoring Engine	Likelihood (1-5) * Impact (1-5) dynamically sets score & severity level	PASS ✓
SoA Immutable Snapshots	Creating SoA snapshot locks entries into immutable JSON payload	PASS ✓
Viewer Role Authorization	Viewer role attempting asset/risk mutation returns HTTP 403 Forbidden	PASS ✓
Cross-Tenant IDOR Attack	Direct ID manipulation in URL returns HTTP 404 (Resource Not Found)	PASS ✓
Nested Form Action Integrity	Updating a risk never invokes the delete action handler	PASS ✓
CSRF & Safe Session Timeout	Expired tokens redirect gracefully with branded 419 error page	PASS ✓
Multi-Format Exporters	CSV, PDF (DomPDF), and Excel (Maatwebsite) generate without warnings	PASS ✓

13. Future Roadmap, REST API Strategy & References

13.1 Phase 8B / Phase 9 Architectural Roadmap

- **Headless REST API (Phase 8B):** Deploy Laravel Sanctum tokenized endpoints to support CI/CD evidence pipelines (automated GitHub/AWS scanner telemetry into control responses).
- **Dedicated Audit Logging UI:** Visual audit log interface leveraging Owen-it Laravel Auditing to trace model mutations with before/after diffs.
- **Third-Party Webhooks:** Automated Slack, Microsoft Teams, and email digest broadcasts for upcoming audit milestones.

14. References & Academic Bibliography

- **[1] International Organization for Standardization:** ISO/IEC 27001:2022 — Information security, cybersecurity and privacy protection — Information security management systems — Requirements. https://www.iso.org/standard/27001
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